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Department of Mathematics
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Math 201 Major Exam I Term 251

Original exam date: 1 October 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The parametric curve

$$x = 2t^2 - 8t, \qquad y = 3t^4 - 4t^3$$

has

(a) a horizontal tangent line at the point $(6, -1)$

(b) a vertical tangent line at the point $(0, 0)$

(c) a horizontal tangent line at the point $(-8, 16)$

(d) a horizontal tangent line at the point $(-6, -1)$

(e) a vertical tangent line at the point $(2, 0)$

2. The number of intersection points between the two polar curves

$$r = 1 - 2\cos\theta \quad \text{and} \quad r = 3\sin(2\theta)$$

is

(a) 7

(b) 11

(c) 13

(d) 9

(e) 17

3. If the points

$$P(1, -2, -k), \quad Q(k-1, 1, 0), \quad R(9, 10, 3k)$$

are collinear, then $(2k-1)^2 =$

(a) 36

(b) 64

(c) 81

(d) 49

(e) 90

4. The area of the triangle with vertices $P(0,0,2)$, $Q(1,0,3)$, $R(-3,2,0)$ is equal to

(a) $\dfrac{3\sqrt{7}}{2}$

(b) $\dfrac{5}{2}$

(c) $\dfrac{3}{2}$

(d) $\dfrac{\sqrt{11}}{2}$

(e) $\dfrac{\sqrt{80}}{2}$

5. The volume of the parallelepiped with adjacent edges $\vec{\mathbf{u}}, \vec{\mathbf{v}}$ and $\vec{\mathbf{w}}$, where

$$\vec{\mathbf{u}} = \langle 1,2,1\rangle \quad \text{and} \quad \vec{\mathbf{v}} \times \vec{\mathbf{w}} = \langle -24,-24,24\rangle,$$

is equal to

(a) 36

(b) 48

(c) 12

(d) 0

(e) 24

6. The parametric curve

$$x = \frac{t^3}{3}, \qquad y = 1 + t^2 + \frac{t^4}{8}$$

is

(a) concave upward for $-1 < t < 0$

(b) concave upward for $0 < t < 1$

(c) concave upward for $t < -2$

(d) concave downward for $t > 2$

(e) concave downward for $t > 3$

7. If $\vec{\mathbf{u}} = \langle -1, 2, 2 \rangle$ and $\vec{\mathbf{v}} = \langle 0, 1, 2 \rangle$, then

$$\left\| \frac{\vec{\mathbf{u}} \times \vec{\mathbf{v}}}{\|3\vec{\mathbf{v}} - \vec{\mathbf{u}}\|} \right\| =$$

(a) $\dfrac{1}{\sqrt{6}}$

(b) $\dfrac{1}{\sqrt{2}}$

(c) $\dfrac{1}{2}$

(d) $\dfrac{2}{3\sqrt{2}}$

(e) $\sqrt{6}$

8. The slope of the tangent line to the polar curve

$$r = 2 + 3\sin\theta$$

at the point corresponding to $\theta = \pi$ is equal to

(a) $-\dfrac{2}{3}$

(b) -1

(c) $-\dfrac{5}{2}$

(d) $\dfrac{1}{2}$

(e) 0

9. The polar equation

$$r = \frac{2}{1 + \cos\theta}$$

can be converted to the rectangular equation

(a) $x^2 + x + y^2 = 2$

(b) $y = \dfrac{2}{1 + x}$

(c) $x = 2$

(d) $x^2 + y^2 + 2x = 1$

(e) $y^2 = 4 - 4x$

10. Which of the following sets of parametric equations describe a parabola or any part of a parabola?

(a) $x = 2t^2,\ y = t^4 + t$

(b) $x = e^{2t},\ y = e^t$

(c) $x = e^{-t},\ y = e^{2t} - 1$

(d) $x = \tan^2 t,\ y = \sec^2 t$

(e) $x = 3\cos t,\ y = 7\sin t$

11. The projection of $\vec{\mathbf{u}} = \langle 5, -1, -1\rangle$ onto $\vec{\mathbf{v}} = \langle -1, 5, 8\rangle$ is

(a) $\left\langle 2,\ -\frac{3}{8},\ 4 \right\rangle$

(b) $\left\langle \frac{2}{3},\ -\frac{2}{5},\ \frac{1}{8} \right\rangle$

(c) $\left\langle -1,\ 1,\ -\frac{5}{8} \right\rangle$

(d) $\left\langle \frac{1}{3},\ -\frac{1}{5},\ \frac{1}{8} \right\rangle$

(e) $\left\langle \frac{1}{5},\ -1,\ -\frac{8}{5} \right\rangle$

12. In the xy-plane, if the vector $\vec{\mathbf{u}}$ is of magnitude 4 and makes an angle of $\frac{\pi}{6}$ with the positive x-axis, and the vector $\vec{\mathbf{u}} + \vec{\mathbf{v}}$ is of magnitude 6 and makes an angle of $\frac{2\pi}{3}$ with the positive x axis, then

(a) $\vec{\mathbf{v}} = \langle -3 - \sqrt{3},\ \sqrt{3} - 2\rangle$

(b) $\vec{\mathbf{v}} = \langle -2 - 3\sqrt{3},\ 2\sqrt{3} - 3\rangle$

(c) $\vec{\mathbf{v}} = \langle 2,\ \sqrt{3}\rangle$

(d) $\vec{\mathbf{v}} = \langle 2\sqrt{3},\ -2\sqrt{3}\rangle$

(e) $\vec{\mathbf{v}} = \langle -3 - 2\sqrt{3},\ 3\sqrt{3} - 2\rangle$

13. The area of the region inside the polar curve $r = 2\sin\theta$ and outside $r = 2(1 + \cos\theta)$ is equal to

(a) $4 - \pi$

(b) $\frac{2\pi - 1}{8}$

(c) $2 - \frac{\pi}{2}$

(d) $2\pi - 5$

(e) $\pi - 3$

14. If $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ are two unit vectors in 3D-space and

$$\vec{\mathbf{a}} = \text{Proj}_{\mathbf{v}}\,\mathbf{u}, \qquad \vec{\mathbf{b}} = \text{Proj}_{\mathbf{u}}\,\mathbf{v},$$

then $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} =$

(a) $(\vec{\mathbf{u}} \cdot \vec{\mathbf{v}})^2$

(b) $(\vec{\mathbf{u}} \cdot \vec{\mathbf{v}})^3$

(c) 0

(d) 1

(e) $(\vec{\mathbf{u}} + \vec{\mathbf{v}}) \cdot (\vec{\mathbf{u}} - \vec{\mathbf{v}})$

15. The area of one petal of the rose curve $r = 2\cos(3\theta)$ is equal to

(a) $\dfrac{3\pi}{7}$

(b) $\dfrac{2\pi}{3}$

(c) $\dfrac{\pi}{6}$

(d) $\dfrac{\pi}{3}$

(e) $\dfrac{3\pi}{4}$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$y'(t) = 12t^2(t-1) = 0$ at $t = 1$: horizontal tangent at $(-6, -1)$.
2	**(d)**	$4 + 4$ non-pole solutions (both representations) plus the pole: 9 points.
3	**(d)**	$\overrightarrow{PR} = 4\overrightarrow{PQ} \Rightarrow k = 4$, so $(2k-1)^2 = 49$.
4	**(c)**	$\overrightarrow{PQ} \times \overrightarrow{PR} = \langle -2, -1, 2\rangle$, norm 3, area $= \frac{3}{2}$.
5	**(b)**	$V = \lvert\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})\rvert = \lvert -24 - 48 + 24\rvert = 48$.
6	**(c)**	$y'' = \frac{t^2-4}{2t^4} > 0$ for $\lvert t\rvert > 2$: concave upward for $t < -2$.
7	**(b)**	$\lVert\mathbf{u} \times \mathbf{v}\rVert = 3$, $\lVert 3\mathbf{v} - \mathbf{u}\rVert = 3\sqrt{2}$, ratio $\frac{1}{\sqrt{2}}$.
8	**(a)**	$\frac{dy}{dx} = \frac{r' \sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta} = \frac{-2}{3}$ at $\theta = \pi$.
9	**(e)**	$r + x = 2 \Rightarrow x^2 + y^2 = (2-x)^2 \Rightarrow y^2 = 4 - 4x$.
10	**(b)**	$y = e^t \Rightarrow x = y^2$ with $y > 0$: half of a parabola; others are not.
11	**(e)**	$\frac{\mathbf{u}\cdot\mathbf{v}}{\lVert\mathbf{v}\rVert^2}\mathbf{v} = -\frac{1}{5}\mathbf{v} = \langle \frac{1}{5}, -1, -\frac{8}{5}\rangle$.
12	**(e)**	$\vec{\mathbf{u}} = \langle 2\sqrt{3}, 2\rangle$, $\vec{\mathbf{u}} + \vec{\mathbf{v}} = \langle -3, 3\sqrt{3}\rangle$, subtract.
13	**(a)**	$A = \frac{1}{2}\int_{\pi/2}^{\pi} \left[4\sin^2\theta - 4(1+\cos\theta)^2\right] d\theta = 4 - \pi$.
14	**(b)**	$\mathbf{a} = (\mathbf{u}\cdot\mathbf{v})\mathbf{v}$, $\mathbf{b} = (\mathbf{u}\cdot\mathbf{v})\mathbf{u}$, so $\mathbf{a}\cdot\mathbf{b} = (\mathbf{u}\cdot\mathbf{v})^3$.
15	**(d)**	$A = \frac{1}{2}\int_{-\pi/6}^{\pi/6} 4\cos^2 3\theta \, d\theta = \frac{\pi}{3}$.